

Student Answer Sheet

Subject: Artificial Intelligence

Definition:

Artificial Intelligence (AI) is the simulation of human intelligence processes by machines, especially computer systems. These processes include learning, reasoning, and self-correction. AI enables machines to perform tasks that typically require human intelligence, such as visual perception, speech recognition, and decision-making.

Body:

AI applications span multiple domains. Machine Learning (ML) is a subset of AI where systems learn from data without being explicitly programmed. Deep Learning uses neural networks with many layers to recognise patterns. Natural Language Processing (NLP) enables machines to understand and generate human language. Computer Vision allows machines to interpret visual information. Key techniques include supervised learning, unsupervised learning, and reinforcement learning. AI is used in healthcare for diagnostics, in finance for fraud detection, and in transportation for self-driving cars.

Conclusion:

Artificial Intelligence represents a transformative technology reshaping industries. While AI offers immense benefits in automation, efficiency, and problem-solving, it raises ethical concerns about privacy, bias, job displacement, and autonomous decision-making. Responsible AI requires transparency, fairness, and accountability.